MEBRIE AWOKE YIZNGAW

AI/ML Engineer | Software Engineering

mebrieawoke941@gmail.com

+251922545447

Arat kilo, Addis Ababa, Ethiopia

Mebrie Awoke

Mebrie Awoke



SUMMARY

Self-taught software developer and fourth-year Information Science student at Addis Ababa University, focused on Artificial Intelligence, Machine Learning, and intelligent software systems. Experienced in backend development, full-stack application development, and building AI-powered solutions.
Interested in developing practical AI/ML solutions involving natural language processing, computer vision, predictive modeling, and intelligent automation. Combines software engineering knowledge with a strong commitment to continuous learning, experimentation, and solving real-world problems through technology. Seeking an opportunity to contribute to AI engineering, applied research, model development, and intelligent system deployment.

Education

2023 – Present
Addis Ababa , Ethiopia

University Student- Addis Ababa University
BSc in Information Science Addis Ababa University, Ethiopia
Fourth-Year Student
Self-Taught Software Development & AI/ML
Continuous practical learning through software projects, technical courses, and AI/ML study.

Technical Skill

Programming: Python, JavaScript, SQL, HTML5, CSS3, Java, C++
AI & Machine Learning: PyTorch, scikit-learn, Deep Learning, CNNs, Transfer Learning, ML models like Supervised, Unsupervised and Reinforecement Learning
NLP & Generative AI: Transformers, LLMs, RAG, LangChain, Hugging Face, Semantic Search
Backend: Flask, FastAPI, Django, REST APIs
Frontend: React, React Native
Databases: MongoDB, SQL, ChromaDB
Tools: Git, GitHub, VS Code, Software Testing

Projects

SmartFarm — AI-Integrated Agricultural Management System
Technologies: Python,HTML,CSS,JS, Machine Learning
- Develop an agricultural technology project integrating software engineering with AI-oriented functionality.
- Explore how intelligent systems can support agricultural activities and improve access to useful information.
- Design the application with a focus on practical usability and extensibility.

GitHub: SmartFarm

AI-Powered Chatbots — Adwa Victory, Medibot and Other RAG based Assistants
- Work on AI-driven chatbot solutions designed to provide users with accessible information through conversational interfaces.
- Apply backend development and AI integration concepts to connect conversational functionality with application services.

- Developed a medical chatbot using Retrieval-Augmented Generation (RAG) to answer questions from PDF documents.
- Integrated Hugging Face embeddings and ChromaDB for semantic search and relevant document retrieval.
- Used LangChain and Groq LLMs to generate context-aware responses.
- Github: AdwaAI Medibot

RAG-Based Telegram Chatbot
Technologies: Python, Retrieval-Augmented Generation, Groq API, Telegram Bot API
- Develop a conversational application that integrates language-model capabilities into a Telegram interface.
- Explore retrieval-based approaches to providing relevant responses and building practical AI applications.
- Integrate external APIs and backend functionality to support chatbot interactions.

HireLens — AI-Powered Resume and Job Matching System
Technologies: Python,FastAPI, React
- Developed a job-fit web application that compares a candidate's resume and professional profile against a job description to assess job suitability.
- Generates a job-match percentage to help candidates understand how closely their qualifications align with a target role.
- Identifies matching skills and missing skills by analyzing resume content and job requirements.
- Provides personalized recommendations to help candidates identify skill gaps and improve their job applications.
- Applies text processing and AI-oriented analysis to support data-driven career guidance.
- Github: HireLens

Music Recommendation System
Technologies: Python, Machine Learning, Pandas, Scikit-learn
- Developed a recommendation engine that suggests songs based on user preferences.
- Applied machine learning techniques to analyze music data and identify relevant recommendations.
- Personalized song discovery based on user interests and listening preferences.
- Github: SmartMusic

For more projects Visit My Github : Mebrie Awoke

Professional Experience

Arada Subcity Innovation Intern — Addis Ababa, Ethiopia
- Participate in software development and digital service initiatives within a public-sector innovation environment.
- Explore opportunities to improve administrative processes and service delivery through practical software solutions.
- Apply programming, API development, database design, and software engineering concepts to project development.
- Develop practical understanding of technology-driven public service improvement and digital transformation.

Backend Developer

Website & Mobile App Developer paid Intern — Orient PLC
- Worked on web and mobile application development for an e-commerce platform.
- Implemented application features and integrated frontend and backend functionality.
- Worked with modern development tools and APIs.

Full-Stack Freelancer — Upwork
- Developed web applications and software solutions for client requirements.
- Worked across frontend and backend components.
- Used Git/GitHub and modern web-development technologies.

In addition to this, I have:
- Designed and developed personal software projects to solve real-world problems and strengthen my software engineering skills.
- Built and customized software solutions for small businesses, including clinics and shops.

Interests

INTERESTS: Artificial Intelligence and Machine Learning | Deep Learning and Computer Vision | Natural Language Processing and Large Language Models | Generative AI and AI Agents | Applied AI Research and Intelligent Systems Development

CAREER OBJECTIVE

To contribute to the development of practical, reliable, and impactful AI/ML solutions through model development, experimentation, data-driven problem-solving, and software integration while continuing to grow as an AI engineer and applied machine learning practitioner.